

DAY-05

DATE:-9/09/2025

Assignment operators: These are used to assign values to the variables.

Examples:

a = 6

a

6

a += 4

a

10

a -= 3

a

7

a **= 2

a

49

a = a//5

a

9

Identity Operators: Used in loops and conditional statements.

is is not

a = 8

b = 3

a is b

False

a is not b

True

a = 8

b = 8

a is b

True

Membership Operators:

I am the member of your family - **False**

I am **not** a member of your family - **True**

you are a member of your family - **True**

you are **not** a member of your family - **False**

num = [1,8,3,9,0]

1 in num

True

2 in num

False

8 not in num

False

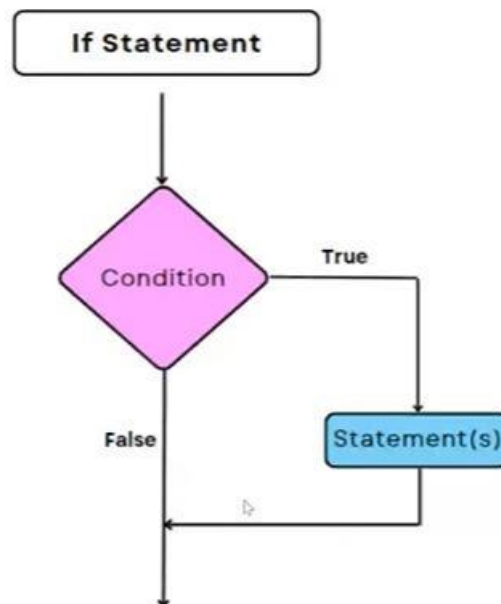
Conditional Statements: It allows us to make decisions in code.

They check conditions(expression in true or false) and execute different blocks of code accordingly.

Types of conditional statements in python:

1. if statement - executes a block only if the condition is true.
2. if.. else statement - provides two paths: one if condition is true
another if false.
3. if....elif..else ladder - Multiple conditions checked one by one
4. Nested if - using one if statement inside another if statement

1. if statement –



Syntax:

```
if(condition):  
    statements
```

Example:

#eligibility checking for election voting in india

```
age = int(input("Enter your age: "))
```

```
country = input("Enter your country : ")
```

```
if(age>=18 and country=="India"):
```

```
    print("eligible for voting")
```

OUTPUT:-

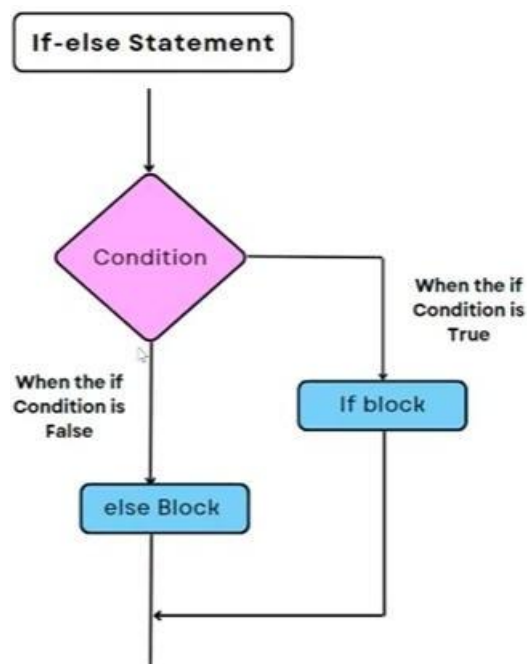
Enter your age: 12

Enter your country : India

Not eligible

In case of developer checking internal conditions it is fine to use if condition but when we are performing any user based operations it is necessary to use if else to avoid dilemma.

2. if.. else statement –



Syntax:

```
if(condition):
```

```
    statements
```

```
else:
```

```
    statements
```

Example:

#eligibility checking for election voting in india

```
age = int(input("Enter your age: "))
```

```
country = input("Enter your country : ")
```

```
country = country.lower()
```

```
print(country)
```

```
if(age>=18 and country=="india"):
```

```
    print("eligible for voting")
```

```
else:
```

```
    print("Not eligibile for voting")
```

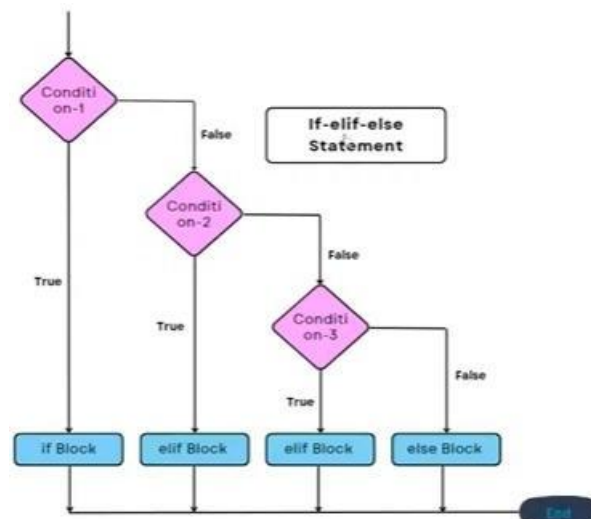
OUTPUT:-

Enter your age: 22

Enter your country : INDIA

Eligible for voting

3. *if...elif...else ladder* –



Syntax:

```
if(condition1):
```

```
    statements of condition1
```

```
else:
```

```
if(condition2):  
    statements of condition2  
  
else:  
    if(condition3):  
        statements of condition3  
    else:  
        else block statements
```

Simplified syntax:

```
if(condition1):  
    statements of condition1  
  
elif(condition2):  
    statements of condition2  
  
elif(condition3):  
    statements of condition3  
  
else:  
    else block statements
```